Text Classification for the UN Sustainable Development Goals: A Comparison of Traditional Models, Transformers and Large Language Models

Vincenzo Uccello 2046148
Sapienza University of Rome
Master's Degree in Engineering in Computer Science and Artificial Intelligence
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uccello.2046148@studenti.uniroma1.it

Abstract

Automatically assigning Sustainable Development Goals (SDGs) to text is useful for analysing research, policy documents and reports, but it is hard: the goals overlap, and existing studies use different datasets and evaluation settings, so their results cannot be compared directly.

This project compares under one protocol several approaches to SDG classification, including a traditional TF-IDF baseline with logistic regression, a fine-tuned RoBERTa model, an existing domain-specific classifier, and three open-weight large language models, evaluated both through prompting alone and after fine-tuning with low-rank adapters. Experiments are conducted on the OSDG Community Dataset, which contains short texts associated with individual goals, and the SDGi Corpus, which contains longer documents that may be associated with multiple goals. The evaluation examines performance within the training domain and the ability of models to generalise across datasets, as well as the difficulty of predicting individual SDGs.

The results show that the best approach depends on the dataset and classification setting. Although transformer models improve on traditional methods in some experiments, simpler approaches remain competitive, particularly when trained on data from the target domain. Fine-tuned large language models show better cross-dataset generalisation than the other evaluated models, whereas prompting alone produces weaker results. The classification of documents associated with multiple goals also remains challenging. For example, sdgBERT is the strongest model on OSDG-CD but no better than an OSDG-trained TF-IDF baseline on SDGi, while the LoRA-fine-tuned Qwen and Gemma lose the least when moving between datasets.

1 Task description/Problem statement

The United Nations Sustainable Development Goals (SDGs) are a set of 17 goals and 169 associated targets adopted in 2015 as part of the 2030 Agenda for Sustainable Development. They address a wide range of social, economic, and environmental challenges and provide a common framework for sustainable development across countries and institutions [23]. As research, policy documents, and sustainability reports increasingly refer to these goals, automatically identifying the SDGs associated with a text has become a relevant task.

The task considered in this report is SDG text classification: given a piece of text, such as a paragraph, a report section, or a complete document, the objective is to determine which SDGs it addresses. Unlike conventional text classification tasks in which categories may be relatively distinct, the SDGs are strongly interconnected, meaning that a text can address issues related to several goals at the same time. For example, a discussion of access to education for disadvantaged communities may be relevant to both SDG 4 (Quality Education) and SDG 10 (Reduced Inequalities). The task is therefore not just identifying explicit references to a goal, but also determining whether the content is relevant to its themes.

Formally, let x denote an input text and let $\mathcal{S} = \{1, \ldots, K\}$ denote the set of SDG labels considered. In the multi-class formulation, a single goal is predicted for each text, whereas in the multi-label formulation, the output is a non-empty subset of $\mathcal{S}$. For score-based classifiers, let $s_k(x)$

Dataset	Input Text (Paraphrased)	Gold	Model Prediction	Error / Behavior Type
OSDG-CD	Paragraph discussing teacher absenteeism and performance-based pay in schools (annotator agreement: 0.20).	SDG 1	SDG 4 (RoBERTa, prob. 0.94)	Label noise / Ambiguous text
SDGi	Excerpt from a local review of Bonn on equal opportunities, diversity, and administration policies.	{10}	{8} (TF-IDF, SDGi-trained)	Semantic overlap (SDG 10 vs. 8)
SDGi	Excerpt on government social programs aimed at eliminating hunger and extreme poverty.	{2}	{1, 2} (TF-IDF, SDGi-trained)	Multi-label partial capture
OSDG-CD	Paragraph arguing that secure property rights are necessary to finance businesses via loan collateral (annotator agreement: $\geq$0.60).	SDG 8	SDG 8 (RoBERTa, prob. 0.77)	Correct prediction

Table 1: Representative examples of inputs, gold labels, model predictions, and corresponding behavioral patterns.

be the score assigned to goal k. In the multi-class setting, the predicted label will be

$$\hat{y} = \arg\max_{k \in \mathcal{S}} s_k(x).$$

In the multi-label setting, the predicted set contains the goals whose scores exceed a decision threshold τ:

$$\hat{Y} = \{k \in \mathcal{S} : s_k(x) \geq \tau\}.$$

For this project, if no goal exceeds the threshold τ (selected using validation data), the highest-scoring goal is retained to ensure that the predicted set is never empty.

The distinction between the two formulations matters here because the two datasets used are built differently. OSDG-CD associates each short excerpt with a single SDG, making it a multi-class task, whereas SDGi allows multiple goals per document and is therefore treated as a multi-label task. Throughout the report, I call the first task *Setting A* and the SDGi task *Setting B*. Further details on the datasets and their preprocessing are provided in Section 3.

1.1 Examples

Table 1 presents four examples from the experiments, selected to illustrate different aspects of SDG classification. The input texts have been summarized for conciseness, while keeping the gold labels and model predictions. The examples show some of the difficulties of SDG classification. The results also depend on the model and the dataset used for training, as performance can change when a model is applied to a different dataset.

1.2 Real-world applications

SDG text classification can help organisations organise and analyse large collections of documents without having to review each one manually. For example, a university could use it to group scientific publications, research projects and course descriptions according to the SDGs they address. Similarly, public agencies could analyse development-aid project descriptions and policy documents to examine which goals are receiving attention and how funding is distributed. The predictions would still need to be checked, especially when a document covers more than one goal, but they could make these collections easier to explore.

The same approach could also be used to study changes over time. By classifying UN General Assembly resolutions from different years, for example, researchers could examine how attention to individual SDGs has changed and identify which goals have become more prominent in international policy discussions.

2 Related work

Keyword and query-based systems.

Early approaches to SDG classification often relied on predefined keywords and Boolean queries, which combine terms using operators such as `AND`, `OR` and `NOT`. One example is Elsevier's SDG mapping, which uses these queries to identify research publications related to each goal [4]. Wulff et al. [24] compared several SDG classification systems across different datasets and types of text, finding substantial differences in sensitivity and specificity. They also showed that combining the outputs of different systems can improve per-

formance compared with using them individually. Kashnitsky et al. [10] examined the methodology behind Elsevier's mapping and compared it with other queries and models using manually labelled data, including an assessment of recall.

OSDG and labelled datasets.

Pukelis et al. [15] introduced the Open Source SDG (OSDG) project, which brings together approaches based on keywords, ontologies and machine learning. OSDG 2.0 later expanded the methodology to support 15 languages [16]. Its Community Dataset (OSDG-CD) contains excerpts labelled through volunteer assessments [14], with each excerpt associated with a single goal and varying levels of annotator agreement.

The SDGi Corpus, introduced by Skrynnyk et al. [21], contains longer documents from national and local sustainability reviews in English, French and Spanish. The corpus supports multiple labels per document.

Supervised classifiers.

Angin et al. [1] fine-tuned RoBERTa-large on OSDG-CD (release 04.2022, 15 goals), keeping only high-agreement examples (agreement of at least 0.6), as I do. They reached a multi-class F1 of 0.92, against 0.91 for BERT-large and 0.88–0.89 for logistic regression and SVM. Another publicly available model is `sdgBERT`, a BERT-based classifier trained on 42,065 labelled excerpts and covering SDGs 1–16. Sadick et al. report weighted precision, recall and F1 scores of around 0.90, while the Hugging Face model card reports an accuracy of 0.90 [19, 20]. However, these scores are difficult to compare directly with the results of this study, since the model was trained on OSDG-CD and possible overlap with the test set cannot be ruled out. The same caution applies to other publicly available models trained on OSDG data, such as `jonas/sdg_classifier_osdg` [9].

Large language models.

Recent studies have investigated whether large language models (LLMs) can classify SDGs through prompting or fine-tuning. Cadeddu et al. [2] compared zero-shot prompting, few-shot prompting and fine-tuning across open-weight and proprietary models for single-label classification. Charalampous et al. [3] compared zero-shot predictions with the Elsevier 2025 keyword mapping and found that LLMs tended to assign more SDG labels than the reference system.

Fine-tuning offers another way to adapt LLMs to SDG classification. Low-Rank Adaptation (LoRA) updates a small set of adapter parameters rather than all model weights, while QLoRA combines this approach with a quantised base model to reduce memory requirements [8]. Most closely related to my setting, Sabatini [18] also used both OSDG-CD and SDGi, focusing on improving single-label BERT classification through Confident Learning and keyword similarity. However, the studies reviewed here do not provide a direct comparison of all these approaches under the same experimental setup, covering both single-label classification of short excerpts and multi-label classification of longer documents, as well as transfer between the two datasets.

3 Datasets and benchmarks

OSDG Community Dataset. OSDG-CD [14] contains short text excerpts, usually three to six sentences, taken from publications, policy documents, reports, and abstracts. Each excerpt is associated with one suggested SDG, which volunteers on the OSDG Community Platform are asked to accept or reject. Between three and nine volunteers assess each suggestion, and the dataset records the positive and negative votes together with an agreement score defined as

$$a = \frac{|p - n|}{p + n},$$

where p and n denote the numbers of positive and negative votes, respectively.

I use the release dated 1 October 2023, which contains 42,355 rows and seven columns. After removing examples with more negative than positive votes, 34,933 excerpts remain, with no empty texts, duplicates or conflicting labels. The excerpts contain approximately 96 words on average and cover SDGs 1–16, as SDG 17 is absent. Since volunteers assess only the suggested goal rather than all potentially relevant goals, each excerpt has a single label and the dataset is therefore used for multi-class classification, even though a text may in reality address more than one goal.

For the main experiments, I retain examples with an agreement score of at least 0.6, resulting in 15,984 training, 3,465 validation, and 3,566 test excerpts. The split is created using stratified group k-fold cross-validation with 20 folds, grouping excerpts by their source document and stratifying by SDG. Three folds are assigned to testing, three

to validation, and the remaining folds to training, preventing excerpts from the same source document from appearing in different splits. I also evaluate on the full test split across all agreement levels, containing 5,374 excerpts for the analysis of annotation quality.

SDGi Corpus.

The SDGi Corpus [21] was created by the United Nations Development Programme (UNDP) using Voluntary National Reviews and Voluntary Local Reviews published between 2016 and 2023. It contains 7,350 examples in English, Spanish and French. Although it is considerably smaller than OSDG-CD, its texts are much longer, with an average of around 1,218 words in the processed data, compared with 96 words for OSDG-CD. The dataset was built by extracting pages from the reports that were clearly related to one or more SDGs and removing unrelated content. Its annotation also differs from OSDG-CD. Labels are based on the goals associated with each section in the original reports, rather than being assigned and validated by volunteers. This allows each example to have multiple labels and the corpus covers all 17 goals, although approximately 89% of the examples have only one label.

The official split contains 5,880 training examples and 1,470 test examples, with no separate validation set. The test set is not a purely random sample, as it includes a larger proportion of examples that a preliminary classifier had difficulty classifying [22].

Preprocessing and common label space. The datasets and models do not all cover the same goals, in particular SDGi covers 17 goals, whereas OSDG-CD and sdgBERT cover only SDGs 1–16, so I restrict the experiments to SDGs 1–16, giving a common label space of $K = 16$. SDG 17 is removed from SDGi, and documents left without any label in the common set are discarded.

Following the SDGi dataset's preprocessing recommendations [22], I removed documents containing corrupted text or invalid character sequences. Since the official split has no validation set, I split the English training documents by source file, obtaining 3,587 training and 633 validation documents. The resulting test set contains 965 English documents and 1,342 documents across all three languages.

Table 2 summarises the dataset sizes after preprocessing.

	Train	Dev	Test
OSDG-CD ($a \geq 0.6$)	15,984	3,465	3,566
OSDG-CD (test, all agreement levels)	–	–	5,374
SDGi, official split, 17 SDGs	5,880	–	1,470
SDGi, after text cleaning, 17 SDGs	5,849	–	1,442
SDGi, English only, 17 SDGs	3,587	633	1,041
SDGi, English only, 16 SDGs	3,387	600	965
SDGi, all languages, 16 SDGs	–	–	1,342

Table 2: Dataset sizes after preprocessing. OSDG-CD splits and the SDGi validation split were created for this project. Average text length is 96 words for OSDG-CD and 1,218 words for SDGi. The proportion of multi-label examples in SDGi is 11.5% in training, 11.7% in validation, and 12.1% in the English test set.

4 Existing tools, libraries, papers with code

Several public resources are available for SDG classification. The OSDG tool and its API can automatically assign SDG labels to texts and provide an overview of their relevance to different goals [13]. On Hugging Face, `sadickam/sdgBERT` and `jonas/sdg_classifier_osdg` provide ready-to-use classifiers [19, 9], while the SDGi Corpus is accompanied by a repository containing benchmark code [22]. In my experiments, I use `sdgBERT` as an existing SDG-specific classifier without further training, excluding the `jonas` model because it covers only the first 15 goals and cannot predict SDG 16, which would make the comparison inconsistent with the common set of 16 labels.

For the experiments, I use scikit-learn for the TF-IDF and logistic regression baseline, and the Hugging Face Transformers and Datasets libraries to load and run the RoBERTa encoder. The language models evaluated are Llama-3.1-8B-Instruct, Qwen3.5-9B, and Gemma-4-12B [6, 17, 5]. I fine-tune them using LoRA with Unsloth [7], while LM Studio is used to run the base models locally for zero-shot and few-shot prompting [12].

5 State-of-the-art evaluation

SDG classification studies do not share an evaluation procedure. Keyword-based systems are checked against manually labelled reference sets, with precision and recall [10] or with sensitivity and specificity, including on texts that have nothing to do with the SDGs [24]. Released classifiers report results on their own validation or test splits [19, 9]. Such scores are hard to compare, and a model whose training data overlap with a new test

split can score too well.

The studies on large language models differ again. Cadeddu et al. [2] score single-label predictions with macro-F1, for proprietary and open models used zero-shot, few-shot and fine-tuned. Charalampous et al. [3] use no gold labels: they compare how many goals five zero-shot LLMs assign to 5,000 OSDG-CD texts, and how those goals are distributed, with the Elsevier 2025 keyword mapping as a reference, and find that the models assign 1.15 to 1.81 times as many.

For multi-label classification, the usual metrics are micro- and macro-averaged precision, recall and F1, subset accuracy and Hamming loss. Micro-averaging favours frequent goals. Macro-averaging weights every goal equally, which suits rare goals but is unstable when they have few examples, and different definitions of macro-F1 give different values. The SDGi paper reports macro-F1 for its baselines [21]. Because these studies differ in data, labels and procedure, their scores cannot be compared directly, which is why I test the models on both single-label paragraphs and multi-label documents, including a cross-dataset test from OSDG-CD to SDGi.

6 Comparative evaluation

This project compares different approaches to SDG classification, ranging from a traditional TF-IDF and logistic regression baseline to fine-tuned transformers and large language models. I evaluate TF-IDF and RoBERTa models trained on both datasets, while the three LoRA-fine-tuned LLMs are trained on OSDG-CD only. I also include sdgBERT, an existing SDG classifier used as released, and evaluate the same three LLM families without further training, relying on prompts alone.

The experiments cover two settings with different classification tasks. In Setting A, each OSDG-CD paragraph is assigned a single SDG, whereas in Setting B, each SDGi document can be associated with multiple goals. Training on OSDG-CD and testing on SDGi therefore allows us to examine how well models generalise to a different dataset, where texts are longer and the label structure is no longer the same.

The experiments address four questions. First, do fine-tuned transformers and LLMs improve on the TF-IDF baseline, and by how much? Second, does an existing SDG-specific classifier perform better than general-purpose models, particularly on data it has not seen? Third, can prompting a general-purpose LLM compete with fine-tuning, and what does fine-tuning add? Fourth, where do the models fail: which goals are hardest, how label noise in OSDG-CD affects the results, how errors relate to the content of the texts and to goals that occur together, and how performance changes between languages.

Systems and baselines.

TF-IDF with logistic regression serves as the traditional baseline. TF-IDF represents each text as a vector of weighted word unigrams and bigrams, using a minimum document frequency of 2. Logistic regression learns how these features relate to the SDG labels and produces a score for each of the 16 goals. In Setting A, it performs multiclass classification, so I select the SDG with the highest predicted probability. The resulting OSDG-CD vocabulary contains 154,636 features. I select the regularisation parameter C on the validation set using macro-F1, testing values in $\{1, 3, 10, 30, 100, 300, 1000\}$. The best value, $C = 1000$, lies at the edge of the grid, although the improvement over $C = 100$ is small, with validation macro-F1 increasing from 0.841 to 0.844. For Setting B, where a document can have several goals, I use a one-vs-rest classifier, that is, one binary logistic regression per goal, so that several goals can pass the decision threshold τ (note that the highest-scoring goal is always kept). I tune C and τ on the validation set to maximise micro-F1 and cap the vocabulary at 200,000 features, since the longer SDGi documents produce a much larger one. The selected values are $C = 30$ and $\tau = 0.17$.

RoBERTa-base [11] is fine-tuned separately for the two settings using AdamW, with a learning rate of $2 \cdot 10^{-5}$, batch size 16, weight decay of 0.01, 10% warm-up, gradient clipping at 1.0 and a maximum input length of 256 tokens. Training runs for up to four epochs, with early stopping based on performance on the validation set, using cross-entropy loss for Setting A and binary cross-entropy for Setting B. I run three seeds (42, 43 and 44) for each setting and report the mean and standard deviation. Since SDGi documents are substantially longer than OSDG-CD paragraphs, they are split into chunks of 150 words, with at most 16 chunks per document, and assign each chunk the labels of its original document. This introduces some noise into the training labels, as a chunk may

not contain evidence for every goal assigned to the full document. At evaluation time, chunk scores are combined into document-level predictions using the first chunk, the maximum or the mean, with the aggregation method selected on the validation set.

sdgBERT [19, 20] is evaluated in its original form, without any additional fine-tuning, using its predictions for the 16 SDGs. Since it was trained on OSDG data, I cannot rule out overlap between its training examples and my OSDG-CD test set, which could inflate its results. I therefore include its OSDG-CD scores for reference but avoid comparing them directly with those of models trained on my own split. Its performance on SDGi is more useful for assessing how well it handles texts from a different corpus.

LoRA-fine-tuned LLMs are included to examine whether adapting larger general-purpose models can improve SDG classification. I fine-tune Llama-3.1-8B-Instruct [6], Qwen3.5-9B [17] and Gemma-4-12B (QAT checkpoint) [5] using low-rank adapters ($r = 16$, $\alpha = 16$, attached to the attention and feed-forward projections; 29.1M–65.6M trainable parameters) on frozen 4-bit base models [8], with Unsloth [7] on a Google Colab NVIDIA L4 GPU. Rather than adding a classification head or generating free text, the prompt maps each goal to a single label word (e.g. *Poverty*, *Health*, *Peace*); applying a softmax over the next-token logits of those 16 words yields the class scores, and training minimises cross-entropy over these 16 logits (AdamW, learning rate $2 \cdot 10^{-4}$, batch size 16, bf16 precision, inputs truncated to 200 words). Each model is trained once for one epoch on 6,000 OSDG-CD training paragraphs, and the resulting unmerged checkpoints also score the SDGi chunks for cross-dataset transfer. The same label-word scoring is applied to the untrained base models on the 496-paragraph subsample, which gives a baseline measured in exactly the same way as the fine-tuned models. Unlike RoBERTa, which uses the full 15,984-paragraph split, these models use a smaller subset because fine-tuning alone takes 42–71 minutes per model on the L4 (compared with 11 minutes for RoBERTa), in addition to slower inference.

Prompt-only LLMs are evaluated without task-specific training, allowing me to compare prompting with fine-tuning for the same three model families. These prompted runs (through LM studio) generate a JSON answer, whereas the fine-tuned models and the untrained baseline described above are scored from label-word logits, so the two query methods give slightly different untrained results. I selected again Llama-3.1-8B-Instruct, Qwen3.5-9B and Gemma-4-12B-QAT, whose 4-bit quantised versions could be run locally on my laptop with 8 GB of GPU memory. I use LM Studio [12] to run the models, with a context window of 8,192 tokens for Llama and Qwen and 7,168 for Gemma. Temperature is set to 0 and thinking mode is disabled. The prompt asks each model to act as an SDG annotator and return only JSON, provides the names of the 16 goals, and constrains the output through a JSON schema.

In Setting A, each model selects one goal, either directly from the text (zero-shot) or with two high-agreement training examples per SDG (few-shot), giving 32 examples in total, each shortened to at most 100 words. In Setting B, the models are prompted without examples to return one to three goals in order of importance, using only the first 600 words of each document. When a request exceeds the context window, the input is shortened and the request is retried.

Since running LLMs locally is relatively slow, I evaluate them on fixed test subsamples of 496 OSDG-CD paragraphs (31 per SDG) and 600 English SDGi documents, 12.8% of which have multiple gold labels. I also evaluate the other models on these same items, allowing a direct comparison within each subsample.

As an additional comparison, I test ChatGPT, Gemini and DeepSeek on 144 OSDG-CD paragraphs (nine per SDG) drawn from the same subsample used for the local LLMs. I submit the paragraphs in 18 shuffled batches of eight, using a fresh chat for each batch and asking for one SDG prediction per paragraph in JSON format. The aim is to compare the hosted chatbots with the locally run models and other classifiers on the same examples. Since the exact model versions were not recorded, I treat these results as exploratory rather than as a real benchmark.

Measures and protocol.

In Setting A, I select the highest-scoring SDG and report accuracy, macro-F1, macro-precision, macro-recall and top-3 accuracy, which measures how often the correct goal appears among the three highest-scoring predictions.

For Setting B, I report micro- and macro-

F1, micro-precision, micro-recall, Hamming loss, subset accuracy and macro-averaged average precision. For each model, I tune a decision threshold τ on the validation set to maximise micro-F1, while always retaining at least the highest-scoring goal. I also evaluate a top-1-only variant, which keeps just the highest-scoring goal, and report the hit rate for prompted LLMs, measuring the proportion of documents for which at least one predicted goal matches a gold label.

In addition to evaluating models on data from the same dataset they were trained on, I test OSDG-CD-trained models on SDGi to examine how well they generalise to a different corpus. Beyond the overall scores, the evaluation includes several targeted analyses: per-goal F1 and its correlation with training-set size, accuracy across annotator-agreement levels on the unfiltered OSDG-CD test set, and a perturbation test removing top TF-IDF keywords versus random words. I also examine the link between frequent OSDG-CD confusions and SDGi goal co-occurrence, performance on the single-label SDGi subset and cross-lingual results on Spanish and French

6.1 Results

Setting A: OSDG-CD.

Table 3 reports the results on the OSDG-CD test set ($a \geq 0.6$, 3,566 paragraphs). RoBERTa improves on the TF-IDF baseline by 3.2 percentage points in accuracy and 3.7 points in macro-F1. The three LoRA-fine-tuned LLMs achieve similar results, with accuracy ranging from 0.886 to 0.899 and macro-F1 from 0.847 to 0.866, despite being trained on only 6,000 paragraphs for one epoch. These results are based on single runs, so the small differences between the models should not be interpreted as a reliable ranking. As expected, `sdgBERT` obtains the highest scores (0.950 accuracy and 0.933 macro-F1), but possible overlap between its training data and the test set makes this comparison unreliable.

Interestingly, top-3 accuracy is high for all models, ranging from 0.966 to 0.991, suggesting that the correct SDG is usually among the three highest-scoring goals, even when it is not the top prediction. The main difficulty is therefore often choosing the correct goal among a small number of plausible candidates, rather than ranking it among the leading options.

On the 496-paragraph subsample, the trained models other than `sdgBERT` obtain relatively high accuracies, ranging between 0.809 and 0.845.

Prompted and fine-tuned LLMs. The prompted LLMs perform less well, with the best configuration, Gemma in the few-shot setting, reaching 0.786 accuracy and 0.783 macro-F1, remaining below the TF-IDF baseline (0.809 accuracy and 0.802 macro-F1). The three open-weight LLMs behave quite differently when used with prompts alone. In the zero-shot setting, Gemma-4-12B-QAT (12B parameters) performs best, with 0.764 accuracy, followed by Qwen3.5-9B (9B parameters) at 0.694 and Llama-3.1-8B-Instruct (8B parameters) at 0.486. Although the results follow the same order as model size, the models might also differ in their pretraining and instruction tuning, so model size alone cannot explain the difference.

Adding two examples per SDG improves all three models, but by different amounts. Llama's accuracy rises from 0.486 to 0.565 (+7.9 percentage points), while Qwen improves from 0.694 to 0.774 (+8.1 points). Gemma gains less, increasing from 0.764 to 0.786 (+2.2 points), and remains the best prompted model. This is because Gemma already performs reasonably well with the instructions alone, leaving less room for improvement from the examples.

The lower rows of Table 3 contain two untrained results for each LLM, and they are not identical, because the two experiments query the same model in different ways. The zero-shot rows come from prompting the model through LM Studio to generate a JSON answer. The rows marked "zero-shot (label-word logits)" use a different prompt and read the scores of the 16 goals directly from the next-token logits, as for the fine-tuned models (LoRA). For Qwen and Gemma the two methods give almost the same accuracy (a difference of less than two points). For Llama the difference is about ten points (0.486 vs. 0.587), so its untrained performance depends more on how it is queried.

LoRA fine-tuning leads to biggest gains, measured against the untrained label-word logits rows of Table 3, where I use the same label-token logit procedure to score both the original and fine-tuned models. Llama's accuracy increases from 0.587 to 0.845 (+25.8 percentage points), Qwen's from 0.688 to 0.833 (+14.5 points), and Gemma's from 0.750 to 0.841 (+9.1 points). Llama gains the most, partly because it starts from the lowest ac-

Model	Full test ($n = 3{,}566$)			Subsample ($n = 496$)	
	Acc.	Macro-F1	Top-3	Acc.	Macro-F1
TF-IDF + LR	0.862	0.820	0.966	0.809	0.802
RoBERTa-base (3 seeds)	0.893 ± 0.002	0.857 ± 0.004	0.977	0.835	0.828
Qwen3.5-9B LoRA	0.897	0.860	0.985	0.833	0.828
Gemma-4-12B-QAT LoRA	0.886	0.847	0.979	0.841	0.834
Llama-3.1-8B LoRA	0.899	0.866	0.985	0.845	0.840
sdgBERT†	0.950	0.933	0.991	0.919	0.917
Llama-3.1-8B, zero-shot / few-shot	–	–	–	0.486 / 0.565	0.467 / 0.562
Qwen3.5-9B, zero-shot / few-shot	–	–	–	0.694 / 0.774	0.693 / 0.772
Gemma-4-12B-QAT, zero-shot / few-shot	–	–	–	0.764 / 0.786	0.766 / 0.783
Llama-3.1-8B, zero-shot (label-word logits)	–	–	–	0.587	0.601
Qwen3.5-9B, zero-shot (label-word logits)	–	–	–	0.688	0.687
Gemma-4-12B-QAT, zero-shot (label-word logits)	–	–	–	0.750	0.747

Table 3: Setting A on OSDG-CD. The subsample is the fixed stratified set of 496 test paragraphs on which the prompted LLMs were run. The zero-shot and few-shot rows prompt the model through LM Studio to generate a JSON answer. The untrained rows score the base model from label-word logits, as for the LoRA models. †Possible overlap between sdgBERT's training data and the test set.

curacy in this comparison and therefore has more room to improve. After fine-tuning, the three models are much closer to each other, with only 1.2 points separating the highest and lowest accuracy on this subsample.

Hosted chatbot comparison. I also tested ChatGPT, Gemini and DeepSeek on 144 paragraphs drawn from the 496-paragraph subsample, so that they can be compared with the other models on the same items (Table 4). Gemini reached the highest accuracy among the hosted chatbots (0.868), slightly above fine-tuned Llama (0.861). DeepSeek (0.840) was close to RoBERTa (0.843) and fine-tuned Qwen (0.847), and ChatGPT reached 0.819. All three hosted chatbots are clearly ahead of the local zero-shot models (0.500–0.792), with Gemini 7.6 points above the best of them, Gemma. Gemini is also above the best local few-shot setting (Gemma, 0.833), DeepSeek is about level with it and ChatGPT is slightly below. An obvious reason for the gap with the local zero-shot models is that the hosted models are larger than the 8–12B models I ran locally. These results are only indicative, since the chatbot queries were run manually and I did not record the exact model versions.

Annotation quality and keyword dependence.

Table 5 compares accuracy across annotator-agreement levels and after removing predictive keywords. For readability, the table reports only the highest- and lowest-agreement groups, while the full notebook also includes results for all five agreement bins, as well as additional perturbation

Model	Accuracy	Macro-F1
TF-IDF + LR	0.792	0.781
RoBERTa-base	0.843	0.833
Qwen3.5-9B LoRA	0.847	0.837
Gemma-4-12B LoRA	0.854	0.847
Llama-3.1-8B LoRA	**0.861**	**0.855**
sdgBERT†	0.924	0.920
Llama-3.1-8B, 0-shot / few-shot	0.500 / 0.583	0.462 / 0.578
Qwen3.5-9B, 0-shot / few-shot	0.688 / 0.799	0.679 / 0.798
Gemma-4-12B, 0-shot / few-shot	0.792 / **0.833**	0.793 / **0.830**
ChatGPT (web, zero-shot)	0.819	0.820
Gemini (web, zero-shot)	**0.868**	**0.862**
DeepSeek (web, zero-shot)	0.840	0.835

Table 4: Results on 144 OSDG-CD test paragraphs (nine per SDG) drawn from the 496-paragraph subsample. The local prompted models were run in LM Studio, and the hosted chatbots were queried manually. Best value per group in bold, excluding sdgBERT. †Possible overlap between sdgBERT's training data and the test set.

experiments such as the impact of removing random words. (GitHub notebook).

For the six trained or released models, accuracy drops by 29.0 to 39.9 percentage points between the high-agreement ($\geq$ 0.8) and low-agreement ($<$ 0.2) groups. TF-IDF has the largest drop (-39.9 points), while Qwen LoRA has the smallest (-29.0 points). The drops are even larger for the zero-shot LLMs, ranging from 46.3 to 54.6 points on their subsample. This suggests that paragraphs with low annotator agreement are harder to classify, possibly because they are ambiguous or cover several SDGs despite having only one gold label. These results also support my decision to train the main models only on examples with annotator agreement of at least 0.6, reducing the amount of potentially noisy labels in the training

Model	Agreement bins			Perturbation		
	≥ 0.8	< 0.2	Δ (pp)	Orig.	$-$Top 3	Δ (pp)
TF-IDF + LR	0.874	0.475	-39.9	0.850	0.640	-21.0
RoBERTa-base	0.905	0.575	-33.0	0.865	0.770	-9.5
sdgBERT†	0.950	0.625	-32.5	0.935	0.840	-9.5
Qwen3.5-9B LoRA	0.912	0.622	-29.0	0.900	0.845	-5.5
Gemma-4-12B-QAT LoRA	0.908	0.531	-37.7	0.880	0.830	-5.0
Llama-3.1-8B LoRA	0.914	0.603	-31.1	0.885	0.825	-6.0
Llama-3.1-8B (0-shot)*	0.650	0.187	-46.3	0.600	0.455	-14.5
Qwen3.5-9B (0-shot)*	0.827	0.346	-48.1	0.785	0.675	-11.0
Gemma-4-12B-QAT (0-shot)*	0.873	0.327	-54.6	0.870	0.755	-11.5

Table 5: Accuracy by annotator agreement and after keyword removal. Agreement results use the full test set for the first six models ($n = 5{,}374$; high/low bins: 2,353/320) and a stratified subsample of 1,793 paragraphs for zero-shot LLMs. Perturbation results use 200 paragraphs for the first six models and 100 for zero-shot LLMs, comparing original text with removal of the three strongest gold-label TF-IDF terms. Δ denotes the change in percentage points; †Possible test-set overlap. * evaluated on subsamples

data.

Removing the three strongest TF-IDF terms associated with the gold label has a large impact on the model's performances. TF-IDF loses the most accuracy (21 points), which is expected since the removed words were selected based on the terms TF-IDF relies on most. RoBERTa and sdgBERT both lose 9.5 points, while the LoRA-fine-tuned models perform best in this experiment, with drops of only 5.0–6.0 points. Qwen also retains the highest accuracy after keyword removal (0.845). The zero-shot LLMs lose 11.0–14.5 points on their smaller perturbation subset, suggesting that they rely more heavily on explicit topical words when assigning SDGs. Fine-tuned LoRA models perform the best: their smaller drops suggest that fine-tuning helps them make better use of the remaining context when some of the most informative words are missing.

Setting B: SDGi.

Table 6 reports the results on the 965 English SDGi test documents. As expected, the two models trained directly on SDGi perform best overall, since they are the only ones trained on documents with this type of text and multi-label structure. TF-IDF obtains the highest micro-F1 and macro-F1 (0.751 and 0.744), with RoBERTa slightly behind (0.736 and 0.727). The two models differ more in the kind of errors they make than in overall quality. In fact, TF-IDF is more precise (0.770 vs. 0.735), while RoBERTa has slightly higher recall (0.738 vs. 0.733) and reaches the best subset accuracy (0.716 vs. 0.687), meaning that it more often predicts the complete set of gold labels correctly. On single-label documents, their accuracy is almost identical, at around 0.86. The fact that TF-IDF performs so well suggests that word choice is already a strong signal in these long documents whereas one possible reason for RoBERTa's smaller advantage is that it processes the documents in 150-word chunks rather than seeing the full text at once.

The models trained on OSDG-CD transfer less well to SDGi, with micro-F1 ranging from 0.599 to 0.678. This is not surprising, since they were trained on short, single-label paragraphs rather than long documents with multiple goals. The size of the performance drop varies across models. OSDG-trained TF-IDF and RoBERTa are about 14–15 percentage points below the SDGi-trained TF-IDF model, while sdgBERT performs similarly to them (0.599). The LoRA-fine-tuned Qwen and Gemma models lose less, reaching 0.678 and 0.676 micro-F1, respectively. On single-label documents, their accuracy is 0.827 and 0.820, within about four points of the in-domain models and around ten points above OSDG-trained TF-IDF and RoBERTa. Llama transfers less effectively, with 0.640 micro-F1 and 0.771 single-label accuracy.

The LoRA models are generally quite precise (0.77–0.81), but their recall is lower (0.55–0.60), meaning that while the goals they predict are usually relevant, they often miss other goals mentioned in the same document, which may be related to their training on single-label examples. Their subset accuracy (0.675–0.712) is still relatively close to that of the models trained on SDGi, partly because most documents have only one gold label. Qwen and Gemma seem to transfer better to this different type of text due to their pretrained representations.

Comparison on a common subsample.

I also evaluated the supervised models and prompted LLMs on the same 600 English SDGi test documents (Table 7). The supervised results are consistent with those on the full test set, with RoBERTa trained on SDGi achieving the highest exact-match accuracy (0.721) and predicting the most labels per document (1.43) among the supervised models. This suggests that it handles documents with multiple goals relatively well, although a separate evaluation on that subset would be needed in future work to confirm this. By comparison, the OSDG-trained LoRA models predict only around one label per document (1.01–1.09), largely retaining the single-label behaviour

Model	τ	Full English test set ($n = 965$) Micro-F1	Macro-F1	Micro-P	Micro-R	Subset acc.	Top-1 hit rate	Single-label subset ($n = 848$) Accuracy
In-domain (trained on SDGi)								
TF-IDF + LR	0.17	**0.751**	**0.744**	**0.770**	0.733	0.687	**0.866**	**0.863**
RoBERTa-base	0.40	0.736	0.727	0.735	**0.738**	**0.716**	0.857	0.855
Cross-dataset transfer (trained on OSDG-CD)								
TF-IDF + LR	0.14	0.607	0.596	0.653	0.567	0.587	0.735	0.722
RoBERTa-base	0.22	0.605	0.602	0.683	0.543	0.592	0.733	0.728
sdgBERT (external)	0.14	0.599	0.593	0.639	0.564	0.545	0.715	0.706
Qwen3.5-9B LoRA	0.29	**0.678**	**0.679**	0.788	**0.595**	0.690	**0.820**	**0.827**
Gemma-4-12B-QAT LoRA	0.41	0.676	0.669	**0.810**	0.581	**0.712**	**0.820**	0.820
Llama-3.1-8B LoRA	0.42	0.640	0.631	0.772	0.546	0.675	0.774	0.771

Table 6: Setting B results on the English SDGi test set. Decision thresholds τ are tuned on the validation set. Top-1 hit rate is the proportion of all 965 documents for which the highest-scoring goal belongs to the gold label set. Single-label accuracy is computed only on the 848 documents with one gold label.

Model	Exact-match acc.	Hit rate	Macro-F1	Avg. labels
Supervised models (validation-tuned thresholds)				
TF-IDF + LR (OSDG)	0.582	0.788	0.591	1.27
TF-IDF + LR (SDGi)	0.693	**0.900**	**0.745**	1.34
RoBERTa-base (OSDG)	0.578	0.771	0.601	1.16
RoBERTa-base (SDGi)	**0.721**	0.893	0.726	1.43
sdgBERT	0.540	0.788	0.592	1.28
Qwen3.5-9B LoRA (OSDG)	0.687	0.852	0.679	1.09
Gemma-4-12B-QAT LoRA (OSDG)	0.705	0.833	0.669	1.03
Llama-3.1-8B LoRA (OSDG)	0.672	0.785	0.633	1.01
Prompted LLMs (zero-shot, predicting 1–3 goals)				
Llama-3.1-8B	0.050	0.752	0.546	1.99
Qwen3.5-9B	**0.352**	0.950	**0.628**	1.99
Gemma-4-12B-QAT	0.085	**0.957**	0.554	2.65
Prompted LLMs (top-1 goal only)				
Llama-3.1-8B	0.545	0.615	0.501	1.00
Qwen3.5-9B	0.782	0.902	0.734	1.00
Gemma-4-12B-QAT	**0.785**	**0.907**	**0.737**	1.00

Table 7: Setting B on the common subsample of 600 English SDGi test documents. Supervised models use thresholds tuned on the validation set. Prompted LLMs predict either 1–3 goals or only their highest-ranked goal. Exact-match accuracy requires the predicted label set to match the gold set exactly; hit rate requires at least one matching goal.

learned during training. Their exact-match accuracy remains reasonably high (0.672–0.705), partly because most documents in the test set have only one gold label.

The prompted LLMs behave quite differently when allowed to return between one and three goals. Llama and Qwen predict around two labels per document, while Gemma predicts an average of 2.65, which is particularly high given the label distribution of the dataset. As a result, Qwen and Gemma achieve very high hit rates (0.950 and 0.957), meaning that at least one predicted goal is usually correct, but their exact-match accuracy is much lower (0.352 and 0.085), since the additional predictions often do not belong to the gold label set. This does not necessarily mean that the models fail to understand the documents. Long report excerpts can discuss several SDGs, even though the dataset assigns them only the goal under which the passage was originally classified. The models may therefore identify several relevant topics rather than reproduce the narrower annotation, which is also consistent with the findings of Charalampous et al. on LLMs assigning more labels than a keyword-based reference system. [3].

Restricting the models to their highest-ranked goal changes the results considerably, with Gemma and Qwen reaching 0.785 and 0.782 exact-match accuracy, respectively, and hit rates above 0.90, while Llama reaches 0.545. This improvement is partly explained by the fact that about 88% of the documents have only one gold label, making a single prediction much more likely to match the complete gold set. Gemma and Qwen therefore come within roughly nine percentage points of the maximum exact-match accuracy

possible for a one-label predictor given this distribution. These results show that both models can identify the main goal of a document quite reliably without task-specific training, although they do not solve the full multi-label task. A possible next step would be to score the individual goals and apply a threshold, rather than asking the models to generate a fixed number of labels.

Goal difficulty, confusions and co-occurrence.

The difficulty of classifying individual goals varies across models and datasets. In Setting A, the hardest goal is SDG 8 (Decent Work and Economic Growth), which has the lowest F1 for every model, ranging from 0.49 for TF-IDF to 0.79 for `sdgBERT`, with the LoRA models between 0.53 and 0.62. It is followed on average by SDG 10 (Reduced Inequalities) and SDG 12 (Responsible Consumption and Production). The easiest goals are SDG 16 (Peace, Justice and Strong Institutions), SDG 14 (Life Below Water) and SDG 4 (Quality Education), with F1 scores between 0.94 and 1.00 across models. Class size alone does not explain these differences, since SDG 16 is the largest class in the test set (473 paragraphs), while SDG 14 has only 120 examples and is still among the easiest. SDG 8 and SDG 10, with 131 and 147 paragraphs, are among the hardest. A possible explanation is that goals such as ocean protection, education and justice have more distinctive vocabularies, whereas employment, inequality and production are themes that can appear across several SDGs. The complete per-SDG F1 table and the corresponding heatmap for Setting A are available in Section 4.1.2 of my GitHub notebook.

The confusion matrix of RoBERTa (Figure 1) gives a closer look at these errors. SDG 10 is often predicted as SDG 8 (21 paragraphs, 14.3% of SDG 10), while SDG 8 is confused with SDG 9 (14.5%) and SDG 10 (13.7%). Other frequent errors include SDG 1 (No Poverty) being classified as SDG 10 (9.1%) and SDG 12 being classified as SDG 7 (Affordable and Clean Energy, 12.0%). These goals share themes such as employment, income, disadvantaged groups and energy use in production, so some paragraphs can reasonably relate to more than one goal, even though OSDG-CD assigns only one label.

In Setting B, the results depend more strongly on where the models were trained, as shown in Figure 2. The SDGi-trained models are relatively consistent across goals, although their weakest scores often concern goals with fewer test examples. For TF-IDF, these include SDG 7 (42 documents) and SDG 15 (62 documents), while for RoBERTa they are SDG 14 (33 documents) and SDG 5 (46 documents). These scores should be interpreted with some caution because of the small number of examples. Averaging F1 across all models, SDG 10 is the hardest goal (0.57), followed by SDG 14 and SDG 5 (both 0.59), while SDG 4 (0.73), SDG 6 (0.71) and SDG 11 (0.70) are the easiest.

Pair	Themes	Documents
SDG 11 + SDG 13	Cities, Climate action	112
SDG 9 + SDG 11	Industry and infrastructure, Cities	97
SDG 3 + SDG 11	Health, Cities	96
SDG 8 + SDG 11	Work and growth, Cities	84
SDG 10 + SDG 11	Inequalities, Cities	84
SDG 4 + SDG 10	Education, Inequalities	78
SDG 1 + SDG 10	Poverty, Inequalities	70
SDG 8 + SDG 9	Work and growth, Industry	69

Table 8: Most frequent pairs of SDGs assigned to the same document in the SDGi training and validation sets (3,987 documents, 11.5% with more than one goal).

SDG 10 is one of the goals that transfers particularly poorly from OSDG-CD. While the SDGi-trained TF-IDF and RoBERTa models reach F1 scores of 0.81 and 0.75, respectively, the scores drop to 0.30 for OSDG-trained TF-IDF, 0.48 for OSDG-trained RoBERTa and 0.50 for `sdgBERT`. This is unlikely to be due to a lack of examples, as SDG 10 is actually one of the largest goals in the SDGi test set, with 122 documents. A possible explanation is that inequality often appears alongside other topics rather than as the main subject of a document. In the SDGi training and validation sets, for example, SDG 10 frequently occurs together with SDG 11 (84 documents), SDG 4 (78) and SDG 1 (70), as shown in Table 8. Models trained on single-label OSDG paragraphs may therefore find it harder to recognise inequality as a separate goal when it is discussed alongside other themes in longer documents.

SDG 11 (Sustainable Cities and Communities) also stands out in the co-occurrence analysis (Table 8), appearing in five of the eight most frequent pairs, probably because many SDGi documents focus on local issues where urban development is discussed together with topics such as health, employment, climate and inequality. The TF-IDF error analysis reflects this pattern, as the model predicts SDG 11 for 233 documents despite only 172

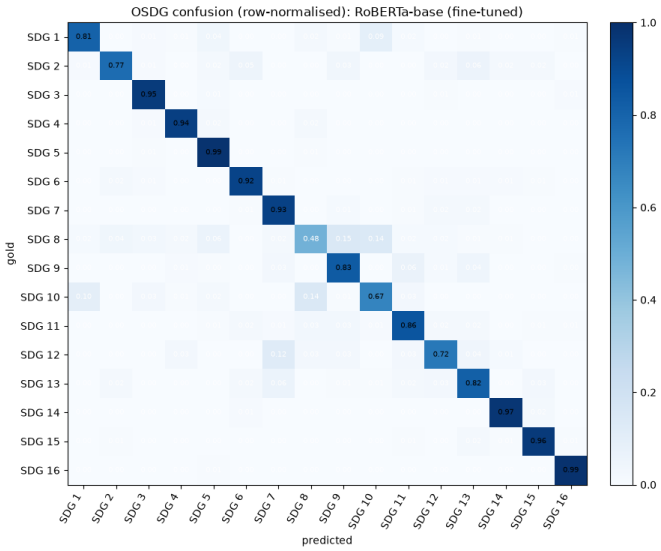


Figure 1: Row-normalised confusion matrix of fine-tuned RoBERTa on the OSDG-CD test set. Rows represent gold labels and columns represent predicted labels.

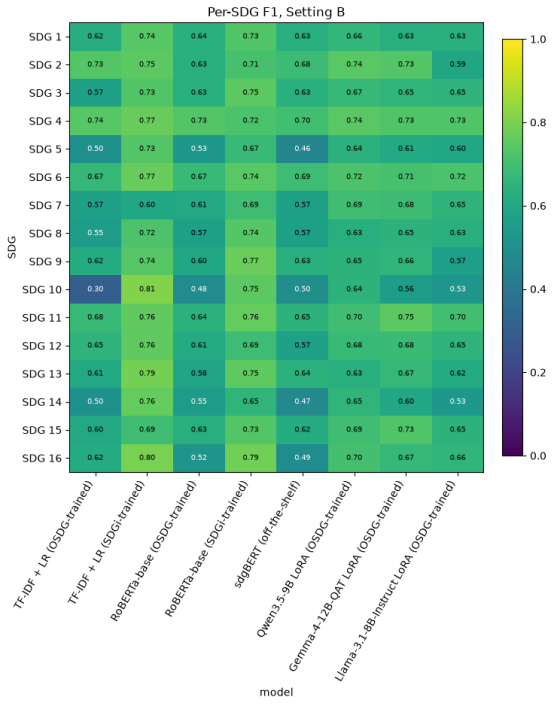


Figure 2: Per-SDG F1 of the evaluated models on the English SDGi test set (Setting B), using validation-tuned thresholds.

having it as a gold label, resulting in high recall (0.89) but lower precision (0.66). SDG 11 also appears frequently in documents where the model makes false predictions for other goals, with 38% of false SDG 3 predictions, 43% of false SDG 9 predictions and 55% of false SDG 13 predictions occurring in documents whose gold labels include SDG 11.

The analysis also highlights differences in how individual goals are misclassified. For example, SDG 11 and, to a lesser extent, SDG 7 are over-predicted (233 predictions for 172 gold labels, and 48 for 42), whereas SDG 15, SDG 2 and SDG 14 are under-predicted. The full per-goal breakdown of false positives, false negatives, precision and recall is available in Table E2 (Section 6.4) of my GitHub notebook.

Overall, the confusions in OSDG-CD and the co-occurrences in SDGi overlap only partly. Three pairs of goals show up in both: SDG 8 and SDG 9, SDG 1 and SDG 10, and SDG 9 and SDG 11. For these pairs, some of the OSDG-CD errors may simply be paragraphs that fit both goals, since each paragraph there has only one label. Other patterns do not match. The largest OSDG-CD confusion, between SDG 8 and SDG 10, is not among the most frequent SDGi pairs, and the most frequent SDGi pair, SDG 11 and SDG 13, is not among the main OSDG-CD confusions. This is probably because the two datasets differ in text type and in how the goals are assigned.

Language analysis.

Table 9 reports micro-F1 on the SDGi test set for each language. Since all models were trained only on English data, the TF-IDF and RoBERTa models lose quite a lot of performance on Spanish and French, with the SDGi-trained versions reaching average micro-F1 scores of 0.514 and 0.554, respectively. OSDG-trained TF-IDF performs even worse, dropping to 0.224. The LoRA-

fine-tuned LLMs handle the language change much better, maintaining scores of 0.630–0.676 on Spanish and reaching 0.869–0.923 on French, in some cases even exceeding their English results. Their multilingual pretraining may help explain this difference, although the non-English results should be treated with caution because the subsets are small and contain different documents. Overall, LoRA-adapted LLMs handle non-English input much better than the other models.

Model	EN (965)	ES (208)	FR (169)	ES+FR (377)
In-domain (trained on English SDGi)				
TF-IDF + LR	**0.751**	0.401	0.654	0.514
RoBERTa-base	0.739	**0.535**	0.578	**0.554**
Cross-dataset transfer (trained on OSDG-CD)				
TF-IDF + LR	0.607	0.170	0.290	0.224
RoBERTa-base	0.620	0.380	0.523	0.444
sdgBERT (external)	0.599	0.232	0.432	0.322
Qwen3.5-9B LoRA	**0.678**	**0.676**	**0.923**	**0.787**
Gemma-4-12B-QAT LoRA	0.676	0.670	0.906	0.776
Llama-3.1-8B LoRA	0.640	0.630	0.869	0.737

Table 9: SDGi test micro-F1 by language. The last column is the weighted average over the 377 non-English documents. RoBERTa rows use seed 42; Table 6 reports the three-seed mean.

6.2 Discussion

Limits of the systems.

The dataset matters about as much as the model. On OSDG-CD, where texts are short and each paragraph has a single label, the trained models achieve fairly similar scores, whereas the picture is different on SDGi, where documents are much longer, can have several labels, and the TF-IDF baseline is just as competitive as the fine-tuned transformer. Each approach also has its own weaknesses: TF-IDF relies heavily on individual words, as shown by the 21-point accuracy drop when its strongest terms are removed, making it fast and competitive yet more sensitive to changes in vocabulary. RoBERTa, on the other hand, can process only up to 256 tokens at a time in my setup, so the longer SDGi documents have to be split into 150-word chunks that inherit the labels of the full document even when some of those goals are not actually discussed in that particular part of the text, adding noise to the training data and partly explaining why RoBERTa does not improve on TF-IDF in Setting B. sdgBERT, the best model on OSDG-CD, is no better than the OSDG-trained TF-IDF and RoBERTa models on SDGi (micro-F1 0.599 vs. 0.607 and 0.605).

The comparison between RoBERTa and the LoRA-fine-tuned LLMs is the one most affected by my compute budget, since I carried out all LLM fine-tuning on a free Google Colab NVIDIA L4 GPU with about 24 GB of memory, which was not enough for full fine-tuning of 9B and 12B models. Because the weights and gradients of a 9B model in 16-bit precision alone would require roughly 36 GB without even counting the optimiser states and activations, I had to use LoRA with 4-bit quantization, keeping only 29.1M–65.6M parameters trainable. Even with this approach, training each model for one epoch on 6,000 paragraphs took 42–71 minutes, whereas RoBERTa was trained on the full 15,984-paragraph training split for up to four epochs in about 11 minutes. The LLMs thus saw less than 40% of the available training data, completed only one epoch and were trained just once. I did not test longer training, so I cannot say whether the LoRA models were already close to their best performance or would improve with more data and epochs. The fact that they reach the level of RoBERTa with this smaller budget is encouraging, but it does not show that they are better.

The LoRA models were also trained only on OSDG-CD because of the same computational constraints, a choice that suited the classification setup used for fine-tuning, where a softmax over the 16 goals and cross-entropy loss match the single-label structure of OSDG-CD. Training on SDGi would instead require a multi-label objective and a different prediction procedure, and its documents are much more expensive to process. Using OSDG-CD also allowed me to compare the LoRA models with the other systems trained on that dataset and then evaluated on SDGi. However, this choice has an the effect that, since they learn from single-label examples, they tend to predict about one goal per SDGi document (1.01–1.09 on average), which means that they often miss additional goals. This is mostly a limitation of the training decisions I took, rather than a limitation of the models themselves.

Prompting alone was less consistent in my experiments, first because the results depend on how the model is queried. Llama, for example, obtains 0.486 accuracy when asked to generate a JSON answer but 0.587 when its label-word logits are used instead. A second issue is that, when asked to return several goals, the models tend to predict more labels than are present in the annotations, producing high hit rates but much lower

exact-match accuracy; restricting them to one goal improves the results considerably, especially for Qwen and Gemma, though this is also helped by the fact that about 87.2% of the SDGi test documents have only one gold label. Finally, the hosted chatbot comparison should be taken only as an initial reference, since it involved just 144 paragraphs and the exact models were not recorded.

Besides the training budget discussed above, there are some limitations in the evaluation itself. I also did not run statistical tests to check whether the differences between models are actually meaningful or depend on the particular test examples. These tests would help understand whether a small improvement is consistent across the test examples or could simply be due to sampling noise, especially for the smaller subsets used for prompted LLMs (496 OSDG-CD paragraphs and 600 SDGi documents), which were run as 4-bit quantised models. The main OSDG-CD results also include only examples with annotator agreement of at least 0.6, both for training and testing, so they do not represent performance on the entire dataset. The evaluation on the unfiltered test set partly addresses this, although low-agreement examples are still excluded from training. Finally, the experiments cover only SDGs 1–16, since SDG 17 is not shared by all the datasets and released classifiers used in this study.

Errors and their causes.

A recurring issue in both datasets is that the gold labels do not always describe every goal that could reasonably be associated with a text. In OSDG-CD, volunteers are asked to accept or reject a suggested goal rather than assign all the goals relevant to an excerpt. This means that a paragraph may be counted as incorrect even when the model predicts a reasonable alternative. The examples in Table 1 illustrate this problem. In the first example, a paragraph about teacher absenteeism and pay has gold label SDG 1, despite an annotator agreement of only 0.20, while RoBERTa predicts SDG 4 with probability 0.94. The prediction can be considered just as reasonable as the original label and the large drop in accuracy on low-agreement paragraphs supports this interpretation, although it is not possible to determine how many errors are caused by ambiguous labels rather than by the models themselves.

A similar problem occurs when goals overlap in their themes. For example, the Bonn excerpt about equal opportunities and diversity is labelled SDG 10 but classified as SDG 8 by TF-IDF. The confusion analysis also shows frequent exchanges between SDG 8, SDG 9 and SDG 10, whose themes can overlap in discussions of employment, economic development and inequality. Since OSDG-CD assigns only one label, these cases are treated as errors even when more than one goal could be perfectly justified by the content.

In SDGi, the problem is slightly different because documents can have several labels, but the labels follow the sections of the original reports and are not independently validated. A passage may discuss several goals even though only one is assigned to it, as in the example about hunger and extreme poverty, where the gold set contains only SDG 2 but TF-IDF predicts both SDG 1 and SDG 2. The model is not necessarily introducing an unrelated goal, but exact-match evaluation still counts the prediction as incorrect. The strong association between SDG 11 and other goals also contributes to errors in this setting, as the SDGi-trained TF-IDF model tends to over-predict it. These examples suggest that some errors are related to the way the datasets define and assign their labels, rather than simply to the ability of a classifier to understand the text.

Language is another important limitation. All the models are trained on English data, but their performance on Spanish and French differs considerably. The TF-IDF and RoBERTa models lose a substantial amount of accuracy, while the LoRA-fine-tuned LLMs retain much better results, probably due to their multilingual pretraining. This does not mean that the models are equally reliable across languages, since the non-English test sets are smaller and contain different documents, but it does show that multilingual llms have a clear advantage for this task.

7 Conclusions

This study compared traditional text classification, fine-tuned language models and open-weight LLMs for SDG classification across two datasets with different characteristics. The results show that no single approach performs best in every setting, and that performance depends strongly on the dataset and the type of classification task.

The four questions posed in the evaluation can now be answered. *Do fine-tuned models improve*

on TF-IDF? On OSDG-CD, RoBERTa and the three LoRA-fine-tuned LLMs achieve similar results, improving on TF-IDF by 2.4–3.7 percentage points in accuracy (0.886–0.899 vs. 0.862). However, on SDGi, TF-IDF trained directly on the target dataset obtains the highest micro- and macro-F1 scores, while RoBERTa achieves the highest subset accuracy. *Does the SDG-specific classifier outperform general-purpose models?* Although `sdgBERT` achieves the highest accuracy on OSDG-CD (0.950), possible overlap with its training data makes this result difficult to compare. On SDGi, it offers no clear advantage, obtaining a micro-F1 of 0.599, below OSDG-trained TF-IDF and RoBERTa (0.607 and 0.605) and LoRA-fine-tuned Qwen and Gemma (0.678 and 0.676). *Can prompting compete with fine-tuning?* Not for the open-weight models tested. On the same 496 OSDG-CD paragraphs, the best few-shot result, obtained by Gemma (0.786), remains below TF-IDF (0.809), while LoRA fine-tuning improves accuracy over the corresponding base models by 9–26 percentage points. The hosted chatbots tested on a smaller sample of 144 paragraphs achieve more competitive results, ranging from 0.819 to 0.868 in accuracy, although these results are only indicative. The LoRA models also perform considerably better on the Spanish and French SDGi subsets, reaching a combined micro-F1 of 0.737–0.787, compared with 0.514–0.554 for SDGi-trained TF-IDF and RoBERTa.

Where do the models struggle? In OSDG-CD, SDG 8 is the most difficult goal, followed by SDG 10 and SDG 12, partly because their themes overlap with those of other goals. Performance also drops considerably on paragraphs with low annotator agreement, with accuracy decreasing by 29–40 percentage points between the highest- and lowest-agreement groups. In SDGi, SDG 10 is particularly difficult for models transferred from OSDG-CD. Another issue is that these transferred models tend to predict too few labels, suggesting that performance on a single-label task does not necessarily translate into reliable multi-label classification. Some errors in both datasets are also likely related to the annotation process, since the assigned labels do not always capture every goal that could reasonably be associated with a text.

Overall, these results show that a model performing well on one dataset does not necessarily perform as well on another, even when the task is the same. Testing SDG classifiers on different datasets and label structures is therefore important to understand how well they perform beyond the data on which they were trained.

Future work.

With more computing resources, I would first repeat the LoRA experiments using the full OSDG-CD training split, more epochs and several random seeds. I would also like to compare LoRA with full fine-tuning, since it was not feasible for the 9B and 12B models with the hardware I own. This would help establish whether the current results are mainly due to the training budget or to the adaptation method itself.

Another interesting extension of this work would be to use the error analysis to make the training more targeted. The confusion matrix in Figure 1 shows that some goals, such as SDG 8, SDG 9 and SDG 10, are particularly difficult to distinguish, while the co-occurrence analysis in Table 8 highlights pairs of goals that frequently appear together, especially those involving SDG 11. These findings could be used to select more informative training examples, focus on difficult pairs of goals or also introduce a contrastive objective that helps distinguish between closely related SDGs. For the multi-label setting, the co-occurrence information could also be used to model relationships between goals rather than treating all labels as completely independent. I would also evaluate the models separately on the multi-label documents (117 of the 965 English test documents) to check whether RoBERTa trained on SDGi really handles several goals better than TF-IDF, since the exact-match scores reported here are dominated by single-label documents.

I would also like to investigate whether some of the annotation noise can be addressed instead of simply removing examples with low agreement. For instance, Sabatini [18] explores Confident Learning and keyword similarity to improve SDG classification, and similar techniques could help identify suspicious labels in OSDG-CD before training itself, potentially allowing more of the available data to be used. On SDGi, I would train the LLMs directly with a multi-label objective and, ideally, use complete documents instead of short chunks. Assigning the labels of an entire document to every chunk can introduce noise when some goals are not discussed in a particular part of the text. Using longer inputs or

a document-level aggregation mechanism could help, although both would require much more memory. Since the prompted LLMs often return more goals than are present in the reference annotations, another approach would be to ask them to score each SDG separately and then use a threshold to decide which goals to keep, rather than asking them to return a fixed number. The threshold could also be tuned separately for each language. This could help reduce over-prediction, which is also observed by Charalampous et al. when comparing LLM predictions with keyword-based annotations [3].

Use of GenAI and LLM tools.

I used Claude as a coding and discussion assistant throughout this project. For the data preparation, it helped me preprocess OSDG-CD and SDGi, apply document-level train/validation/test splits, and understand the structure and fields of the two datasets. We discussed the common label space, which I restricted to SDGs 1–16 because SDG 17 is not shared by all datasets and classifiers. We also discussed the choice of the 0.6 agreement threshold for the main experiments and the use of chunking for long SDGi documents. Claude helped me organise the notebook, including its sections, global variables and utility functions, select the evaluation metrics, review the RoBERTa training settings (learning rate, batch size, weight decay, number of epochs and early stopping), and debug code, including GPU and CUDA issues on my RTX 5060 Laptop GPU with 8 GB of VRAM.

Since this was my first time fine-tuning models with Unsloth, I also used Claude to help write the fine-tuning code for the three open-weight models and understand how LoRA adapters work. The decision to use LoRA rather than full fine-tuning was mine, since full fine-tuning exceeded my available GPU memory. I selected the three models from the LM Studio catalogue, taking the 8 GB VRAM limit into account. Claude also helped me connect the LM Studio API to the notebook and run inference by loading and unloading one model at a time.

I also used ChatGPT as a discussion assistant during the experiments and while writing the report. We discussed the experimental design, technical concepts and evaluation methodology, and I used it to investigate some of the results that were not immediately clear. For example, we discussed why TF-IDF performed so well when trained directly on SDGi, why models transferred from OSDG-CD struggled with the multi-label setting, and why they tended to predict fewer labels than expected. ChatGPT helped me review related work on SDG classification, identify possible research papers and explore additional sources of data. When writing the report, I used it to organise some sections, improve the clarity of the explanations and polish the English. More importantly, it helped me think through possible explanations for the results and understand the reasons behind some of the patterns observed in the experiments. The analysis and final interpretation, however, are mostly my own.

I remained responsible for the implementation, experiments, results, references and final content, and checked the material before including it in the report.

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